

Narrative Contagion at the Sovereign Default Boundary

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July 2026

Abstract

A fixed pessimistic belief distortion has little price relevance for a safe sovereign or for one that already prices default. It matters near the default boundary. This paper places that observation in a sovereign pricing model with narrative diffusion and debt-service feedback. The primitive belief gap is exactly single-peaked. The equilibrium price effect is single-peaked under a derivative restriction on the feedback multiplier, and the restriction holds on the reported scenario grid. Narrative adoption follows from an attention-cost rule in which investors adopt when decision value exceeds an idiosyncratic cost. The prevalence process is continuous in time and the annual pricing model is mapped exactly into monthly simulation. A state-space measurement equation separates media coverage from latent investor prevalence. In the quantitative scenario, ten percentage points of latent prevalence raise the spread by as much as 230 basis points, and seeding raises the two-year default probability by at most 5.3 percentage points. These are scenario outputs. A binding cap sets the marginal narrative effect on the modelled bond to zero. It does not eliminate the narrative or its effects on other markets.

JEL codes: D83, D84, E71, F34, G01, H63.

Keywords: narratives, epidemiology, sovereign debt, default boundary, doom loop, text data.

1 Introduction

On July 26, 2012, the president of the European Central Bank stated that the ECB was ready to do whatever was necessary within its mandate to preserve the euro. The Outright Monetary Transactions program was announced in September and was never activated. Research attributes a large fall in Italian and Spanish short yields to the OMT announcements (Altavilla et al., 2016). The intervention also changed redenomination insurance and the expected policy regime (Krishnamurthy et al., 2018). It was not a pure communication shock.

This paper asks when a portable pessimistic claim can move a sovereign spread. The answer comes from the default boundary. Narrative economics supplies the diffusion idea (Shiller, 2017; Flynn and Sastry, 2024). Sovereign pricing supplies the state dependence. The new object is the interaction between a default-probability curve and a measurable media-diffusion process.

The direct belief gap is $G(z) = \Phi(z + \Delta) - \Phi(z)$. It is the probability mass between the ordinary and pessimistic default thresholds. This gap has one peak at $z = -\Delta/2$ and vanishes in both tails. The equilibrium effect multiplies G by the spread-feedback term. A bounded multiplier does not preserve single-peakedness by itself. Proposition 1 states a derivative condition that does, and the quantitative scenario verifies that condition on its state space.

Adoption has a costly-attention foundation. An investor transmits the narrative when its decision value exceeds an idiosyncratic cost. The resulting adoption probability is the cost distribution evaluated at the price relevance of the narrative. This assumption is testable against a competing rule based on perceived mispricing. Media coverage measures diffusion across outlets. It is linked to latent investor prevalence by a measurement equation rather than equated with it.

When $R_0 < 1$, a small seed decays. When $R_0 > 1$, it grows while fundamentals remain in the susceptible interval. Debt-service feedback can then change the fundamentals path. The instantaneous lower zero-drift root rises with prevalence under a local slope condition. This root is not automatically the basin boundary of the full system. The quantitative section therefore reports the vector field and the seeded path. In that scenario the seeded path crosses zero capacity in month 39, while the matched unseeded path survives.

The quantitative claims are asymmetric. Narratives in this model move prices much more than they move defaults. At the hump’s peak, ten percentage points of prevalence are worth 230 basis points of spread, and a full outbreak generates an excess spread that reaches 3115 basis points before the end: magnitudes in the range of the fundamentals-unexplained spread components documented for the euro periphery by De Grauwe and Ji (2013). The same outbreak adds up to 5.3 percentage points to the two-year default hazard at the baseline feedback, a band of 0.8 to 10.2 across the range of gross financing needs that disciplines the feedback parameter, all inside the zone. For a safe sovereign the addition is 0.20 percentage points. A story taxes vulnerable sovereigns heavily while it circulates, and occasionally, in a narrow band of fundamentals, finishes one off.

The policy experiments change carriers, abandonment, or the modelled spread. One-time removal delays a supercritical outbreak under the local constant-rate approximation. Faster abandonment lowers the reproduction number. A binding cap sets the marginal narrative effect on the capped bond to zero. Other market and political channels remain outside the model. The timing costs reported below are conditional on the scenario parameters.

The model combines the narrative diffusion framework of Flynn and Sastry (2024) with the sovereign default boundary and debt-service feedback (Calvo, 1988; Lorenzoni and Werning, 2019). Text methods supply the observable media process (Gentzkow et al., 2019). The contribution is the default-boundary prediction and its separation of primitive from equilibrium state dependence.

The empirical design uses GDELT as the public media core and licensed full text for historical validation. Scheduled fiscal releases discipline timing. Boundary proximity is predetermined and estimated from the lagged response of spreads to fundamental surprises. OMT is treated as a compound event rather than an instrument for narrative removal.

2 A Sovereign, Its Creditors, and a Story

The threshold-default primitive delivers the direct pricing hump. Equilibrium single-peakedness requires a derivative restriction on the feedback term. Contagion results also require the costly-attention adoption assumption. Global default paths require phase conditions beyond the local comparative statics.

2.1 Fundamentals, default, and pricing

Time is monthly and the pricing horizon is one year. A sovereign's fundamentals θ_t measure debt-service capacity in units of annual resources. Fundamentals mean-revert and are drained by borrowing costs:

$$\theta_{t+1} = \rho_m \theta_t + (1 - \rho_m) \bar{\theta} - \chi_m s_t + \varepsilon_{t+1}, \quad \varepsilon \sim N(0, \sigma_m^2), \quad (1)$$

where s_t is an annualized spread. Default is absorbing once $\theta_t < 0$. The monthly coefficients aggregate exactly to the annual law under a constant spread:

$$\rho_m = \rho^{1/12}, \quad \sigma_m = \sigma \sqrt{\frac{1 - \rho_m^2}{1 - \rho^2}}, \quad \chi_m = \chi \frac{1 - \rho_m}{1 - \rho}. \quad (2)$$

Thus twelve monthly innovations have variance σ^2 , and twelve monthly spread charges cumulate to χs after persistence is accounted for. At the annual horizon, the perceived default probability is

$$p(\theta, s) = \Phi(z(\theta, s)), \quad z(\theta, s) = \frac{-\rho\theta - (1 - \rho)\bar{\theta} + \chi s}{\sigma}. \quad (3)$$

The S-shape of p in θ is the paper's load-bearing object, and it comes free: any default technology in which a shock-buffed state crosses a threshold produces one.

A pessimistic narrative circulates among the sovereign's creditors. Carrying the story means pricing the bond under fundamentals worse by ξ : the deficit is understated, the peg will break. The story is portable and self-contained, which is what lets it spread. Its content is a belief distortion rather than an information signal. With prevalence n_t , the fraction of investor wealth carrying the story, the market-clearing default assessment is the wealth-weighted average

$$\bar{p}(\theta, s, n) = (1 - n) \Phi(z) + n \Phi(z + \Delta), \quad \Delta \equiv \frac{\rho \xi}{\sigma}, \quad (4)$$

so the story enters as a shift of size Δ in the standardized distance to default for the infected. Risk-neutral pricing of one-period debt with recovery $\mathcal{R}$ gives the spread as expected loss, $s = (1 - \mathcal{R})\bar{p}$, and since the spread feeds back into the default assessment through (3), the within-period equilibrium is a fixed point.

Lemma 1 (Pricing equilibrium and the doom-loop multiplier). *If $(1 - \mathcal{R})\chi/(\sigma\sqrt{2\pi}) < 1$, the map $s \mapsto (1 - \mathcal{R})\bar{p}(\theta, s, n)$ is a contraction: for every (θ, n) there is a unique equilibrium spread*

$s^*(\theta, n)$, increasing in n and decreasing in θ . Small shocks to the default assessment are amplified by the multiplier

$$M(\theta, n) = \left[1 - (1 - \mathcal{R}) \frac{\lambda}{\sigma} ((1 - n)\phi(z^*) + n\phi(z^* + \Delta)) \right]^{-1} \geq 1,$$

which reaches 2.8 at the scenario maximum. If the feedback crosses the contraction frontier, multiple pricing fixed points can arise.

The scenario sits inside the uniqueness region with maximal loop gain 0.64.

2.2 The contagion block

Prevalence follows a continuous-time adoption process:

$$\dot{n}_t = \beta(\theta_t, n_t)n_t(1 - n_t) - \gamma n_t. \quad (5)$$

This vector field leaves $[0, 1]$ invariant whenever β and γ are nonnegative. The monthly simulation applies the exact logistic flow over one month while holding the current rates fixed.

Assumption 1 (Costly attention and adoption). *Upon encountering a carrier, investor i draws an adoption cost c_i from a distribution H_c on the nonnegative real line. Adoption occurs when*

$$c_i \leq \lambda \left| \frac{\partial s^*(\theta, n)}{\partial n} \right|.$$

The distribution has no mass at zero and has a strictly positive density on its interior. Encounters occur at rate $\bar{m}A$. Hence

$$\beta(\theta, n) = \bar{m}AH_c \left(\lambda \left| \frac{\partial s^*(\theta, n)}{\partial n} \right| \right).$$

The inequality gives the reduced-form adoption function an attention-cost interpretation. It does not make the distorted belief rational. A competing hypothesis makes adoption depend on perceived mispricing rather than decision value. The two mechanisms need not respond alike when a policy caps one sovereign yield. Under Assumption 1, the reproduction number at zero prevalence is

$$R_0(\theta) = \frac{\bar{m}AH_c(\lambda v(\theta, 0))}{\gamma}, \quad v(\theta, n) = \frac{\partial s^*(\theta, n)}{\partial n}. \quad (6)$$

The reproduction number is high where adoption value is high.

3 The Boundary Theorems

3.1 Where stories are priced

Theorem 1 (Primitive pricing hump). *Let $G(z) = \Phi(z + \Delta) - \Phi(z)$ for $\Delta > 0$. The function G is strictly increasing on $(-\infty, -\Delta/2)$, strictly decreasing on $(-\Delta/2, \infty)$, and converges to zero in*

both tails. Its unique maximum is $2\Phi(\Delta/2) - 1$. Because $z^*(\theta, n)$ is strictly decreasing in θ , the direct belief gap $G(z^*)$ is single-peaked in fundamentals.

The equilibrium price impact is

$$\frac{\partial s^*}{\partial n} = (1 - \mathcal{R}) [\Phi(z^* + \Delta) - \Phi(z^*)] M(\theta, n).$$

Proposition 1 (Sufficient condition for an equilibrium hump). *Put $a = (1 - \mathcal{R})\chi/\sigma$ and $d(z, n) = (1 - n)\phi(z) + n\phi(z + \Delta)$. Define*

$$Q(z, n) = \frac{G'(z)}{G(z)} + \frac{ad'(z, n)}{1 - ad(z, n)}.$$

On any index interval $\mathcal{Z}$, suppose $ad(z, n) < 1$ and $\partial Q(z, n)/\partial z < 0$. The equilibrium price impact has at most one interior peak on the corresponding fundamentals interval. It has exactly one peak if Q takes both signs. In the quantitative scenario, the largest value of $\partial Q/\partial z$ on $z \in [-6, 6]$ and $n \in [0, 1]$ is -0.474 . The computed equilibrium effect is therefore single-peaked on the reported state space. In the omitted tails, G is numerically negligible and M converges to one.

The theorem concerns the direct belief gap. The proposition states the additional restriction required for equilibrium feedback to preserve one peak. Boundedness of the multiplier alone is not sufficient.

3.2 Where stories spread

Under Assumption 1, adoption is an increasing function of the equilibrium price impact. The interval result therefore uses Proposition 1.

Proposition 2 (The susceptible zone). *Suppose Proposition 1 holds and H_c is strictly increasing over the attained values. Then $R_0(\theta)$ is single-peaked with vanishing tails. The susceptible set $Z = \{\theta : R_0(\theta) \geq 1\}$ is an interval. It is nonempty if and only if $\bar{m}AH_c(\lambda v_{\max}) \geq \gamma$. A rise in A expands every superlevel set. A rise in γ contracts it. Other parameter changes alter the height and location of v , so endpoint monotonicity requires additional restrictions.*

In the scenario, reducing amplification to one quarter of baseline shrinks the susceptible interval to width 0.001 in capacity units. At baseline the interval maps into spreads between 314 and 3407 basis points. These are model coordinates, not estimated historical boundaries.

Corollary 1 (Boundary proxy). *For small Δ , the direct belief gap satisfies $G(z) = \Delta\phi(z) + O(\Delta^2)$. The lagged local sensitivity of spreads to scheduled fundamental surprises is therefore a model-based proxy for boundary proximity. It is not a sufficient statistic once the multiplier, narrative bite, or recovery varies across countries.*

3.3 From local drift to global paths

The state (θ_t, n_t) moves through a plane. At zero prevalence, a small seed initially decays where $R_0 < 1$ and grows where $R_0 > 1$. When β is constant, the positive steady state is $1 - \gamma/\beta$. In the model β varies with both state variables, so global behavior must be read from the nullclines and vector field.

Proposition 3 (Instantaneous drift boundary). *Let*

$$D(\theta, n) = (1 - \rho_m)(\bar{\theta} - \theta) - \chi_m s^*(\theta, n).$$

Suppose that for each n in an interval N , the equation $D(\theta, n) = 0$ has a lower root $\hat{\theta}(n) > 0$ with $D_\theta(\hat{\theta}(n), n) > 0$. Then $\hat{\theta}$ is continuously differentiable and

$$\hat{\theta}'(n) = \frac{\chi_m s_n^*(\hat{\theta}(n), n)}{D_\theta(\hat{\theta}(n), n)} > 0.$$

This root is the instantaneous lower drift boundary for the scalar fundamentals equation when n is held fixed. It is not by itself the basin boundary of the two-dimensional system.

Different seeded and unseeded outcomes require global phase restrictions. A forward-invariant default region must have $D \leq -d < 0$. The seeded path must enter that region while the unseeded path remains in a safe invariant set. Under these conditions the seeded path reaches default in finite time and the unseeded path does not. Figure 4 reports the nullclines and vector field for the scenario. Figure 3 reports the simulated path. The seeded path crosses zero capacity in month 39. This is a scenario path result, not a consequence of root monotonicity alone.

4 Policy Experiments

Proposition 4 (Policy effects within the model). *If $R_0 > 1$ is constant while prevalence remains near zero, removal of a fraction f delays return to the original prevalence by*

$$\frac{\log[1/(1-f)]}{\gamma(R_0 - 1)}.$$

A rise in γ lowers R_0 pointwise and lowers the endemic prevalence wherever β is constant. A perfectly credible cap on the modelled bond sets $\partial s^/\partial n = 0$ wherever it binds. Under Assumption 1, adoption through the modelled price-relevance channel then stops in that region. The cap also truncates the feedback from that bond spread to fundamentals.*

The cap result concerns one pricing channel. Narratives may still affect CDS, other maturities, auction quantities, capital flight, bank funding, or political choices. A policy backstop can also change expected recovery and the policy regime. OMT is therefore a compound intervention rather than a pure narrative experiment. The calculations below impose a cap on the modelled spread and hold other channels fixed.

Table 1: Quantitative scenario.

Parameter		Value	Category	Basis
Persistence (annual)	ρ	0.90	scenario	fundamentals persistence
Innovation sd (annual)	σ	0.05	scenario	capacity volatility
Long-run capacity	$\bar{\theta}$	0.15	scenario	capacity scale
Recovery	$\mathcal{R}$	0.60	external	haircut evidence (Cruces and Trebesch, 2013)
Feedback	χ	0.20	scenario	equation (7)
Story bite	ξ	0.03	scenario	$\Delta = 0.54$
Abandonment	γ	0.30/mo	scenario	episode half-life target
Peak reproduction	R_0	4	scenario	diffusion-rate target
Seed	n_0	0.02	scenario	observed onset target
Loop gain (max)		0.64	implied	uniqueness margin, Lemma 1

Notes: recovery is externally anchored. Every other non-implied entry is a scenario input. The empirical section defines the moments required for estimation.

If adoption tracks decision value, a binding cap lowers media diffusion through this channel. If adoption tracks perceived mispricing, coverage can persist while the capped yield is stable. Post-OMT text can describe which pattern occurred. It cannot isolate this mechanism because OMT changed tail insurance, conditionality, and the expected policy regime.

5 Quantitative Magnitudes

5.1 Quantitative scenario

Table 1 lists the scenario inputs. Recovery is anchored to the sovereign haircut evidence in Cruces and Trebesch (2013). The remaining values are not estimates. The annual parameters map into $\sigma_m = 0.0151$ and $\chi_m = 0.0175$ through equation (2). The feedback parameter has the cash-flow decomposition

$$\chi_{it} = \text{RefinancingShare}_{it} \times \text{MarketRatePassThrough}_{it} \times \text{FiscalCapacityConversion}_{it}. \quad (7)$$

The scenario value 0.20 is examined over $[0.10, 0.30]$. Empirical implementation would construct each factor from remaining maturity, the market-rate share of maturing debt, official-loan insulation, cash buffers, and the fiscal response. World Bank QPSD supplies public debt stocks and maturity fields for reporting countries (World Bank, 2026). Debt-office cash-flow schedules supply country detail. The narrative parameters determine scenario dynamics until they are estimated from media diffusion and belief responses.

5.2 The hump, the zone, and the outbreak

Figure 1 draws Proposition 1: the price relevance of the story peaks at 230 basis points per ten points of prevalence near the boundary and dies in both tails, and panel (b) places the susceptible zone in spread space, between 314 and 3407 basis points. Figure 2 draws Proposition 2: the baseline zone spans $[0.001, 0.069]$ in fundamentals. At a quarter of baseline amplification the

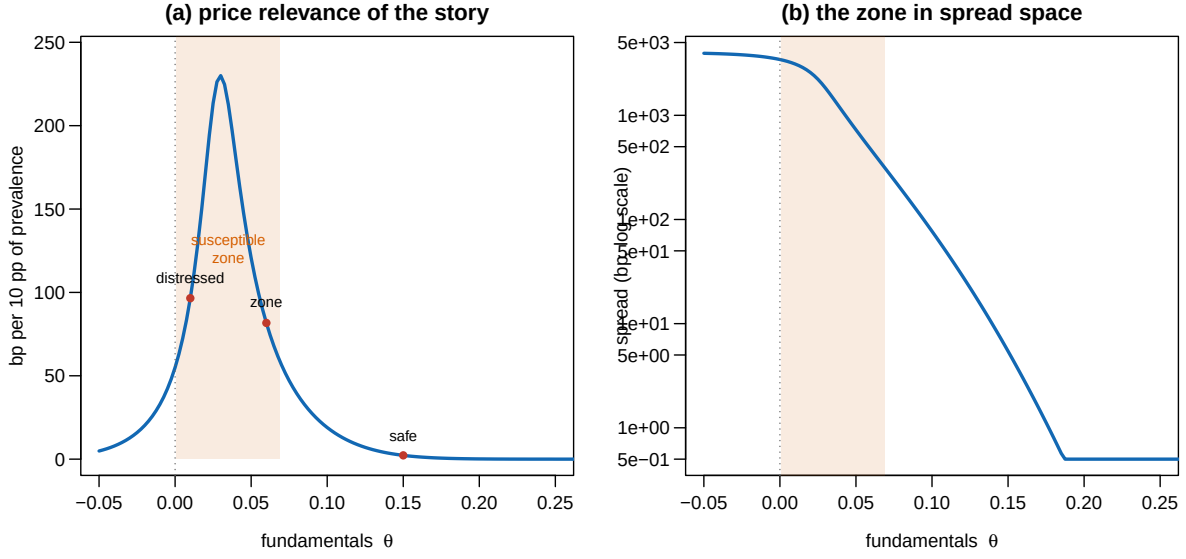


Figure 1: Scenario equilibrium price impact and spread schedule. Shading marks the susceptible interval. Dots mark the paths in Figure 3.

zone degenerates to a point, and at 1.5 times baseline it reaches to $\theta = 0.081$.

Figure 3 reports deterministic scenario paths from the same two percent seed. The safe case starts at $\theta_0 = 0.15$. The low-capacity case starts at 0.01. The susceptible case starts at 0.06 with spread 465 basis points. Its unseeded comparison survives. Its seeded path reaches peak prevalence 72 percent and a peak excess spread of 3115 basis points before capacity crosses zero in month 39.

5.3 The crisis wedge and the policy scorecard

Figure 5 uses eight thousand common-random-number paths per starting point. The seeded minus unseeded default-probability difference reaches 5.3 percentage points at $\theta_0 = 0.04$. It is 0.20 points at $\theta_0 = 0.15$. Raising γ by half lowers the peak difference to 3.0 points. The 400 basis point cap makes the modelled narrative wedge 0.0 in this scenario and also changes the unseeded fundamentals risk. Figure 6 reports conditional timing arithmetic. Moving cap availability from month one to month thirteen raises default probability from 48.5 to 65.4 percent.

Across the feedback scenarios $[0.10, 0.30]$, the peak two-year wedge ranges from 0.8 to 10.2 percentage points. Peak price relevance ranges from 124 to 1931 basis points per ten percentage points of latent prevalence. At the upper value, the maximal loop gain is 0.96. Results near one are sensitive to equilibrium selection and are reported as boundary diagnostics rather than stable quantitative predictions.

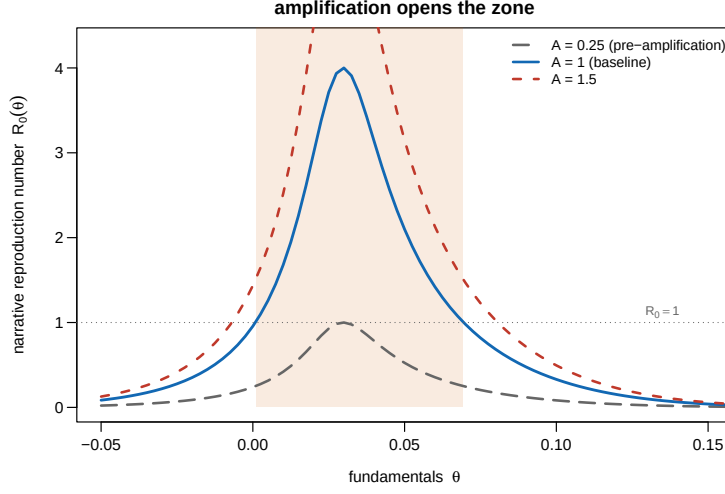


Figure 2: The narrative reproduction number at alternative amplification values. The susceptible interval is the superlevel set above one.

6 Measurement and Empirical Designs

6.1 Text and the latent narrative state

The public core corpus is GDELT 2.0 Global Knowledge Graph and Mentions data, which are downloadable and available through BigQuery (GDELT Project, 2015). Factiva or Lexis-Nexis provides licensed full-text validation for historical episodes. The design does not require private messages, investor chat logs, restricted social-platform archives, or wealth-weighted communication networks.

A narrative is a structured claim. It names an economic state, gives a causal explanation, implies a future sovereign outcome, and carries an action or policy implication. A high-recall dictionary screen selects candidates. A multilingual classifier assigns the structured label. The validation sample is manually double-coded under blinded assignment. The prompt, model version, label rules, and country-language error matrices are frozen before outcome analysis.

Text coverage is not investor wealth prevalence. The measurement equation is

$$\text{TextIndex}_{it} = a_i + \lambda_i n_{it} + e_{it}, \quad n_{i,t+1} = \mathcal{T}(n_{it}, \theta_{it}) + \eta_{it}. \quad (8)$$

A state-space implementation estimates the latent n_{it} and the loading λ_i . A reduced-form implementation uses the text index as a proxy and does not convert coverage changes into percentage points of investor wealth. GDELT timestamps and source identifiers measure media diffusion across outlets. They do not measure person-to-person investor transmission.

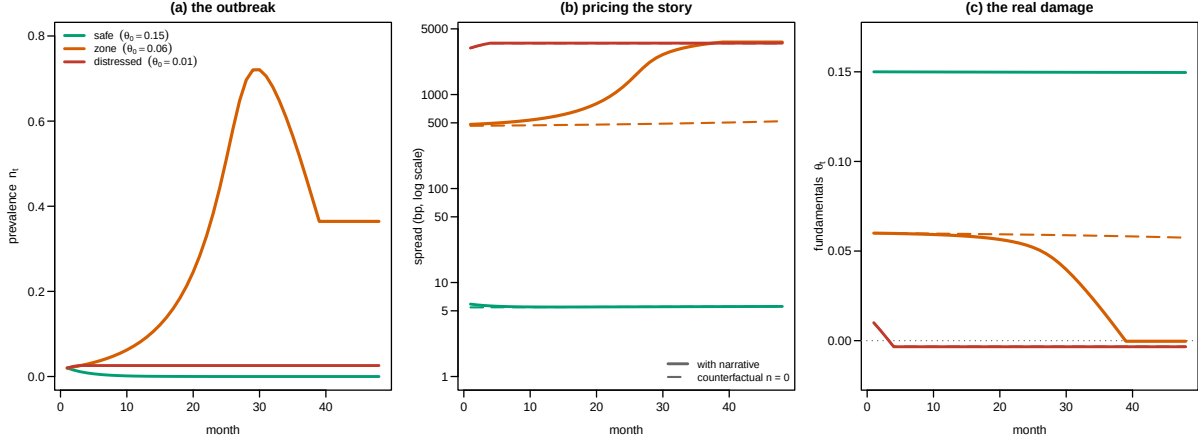


Figure 3: Deterministic paths from a two percent seed. Dashed lines give the $n \equiv 0$ comparison. The susceptible scenario crosses zero capacity at month 39.

6.2 Boundary interaction and release timing

The workhorse local projection is

$$\Delta s_{i,t+h} = \alpha_i + \tau_t + f_h(B_{i,t-1})N_{it} + \Gamma'_h X_{i,t-1} + u_{i,t+h}, \quad (9)$$

where N_{it} is the measured text index or the filtered latent state. The function f_h is estimated by bins or splines. The theory predicts a small effect in both tails and an interior peak. A linear interaction is not the primary specification.

The boundary proxy $B_{i,t-1}$ is estimated before the narrative window. It is the lagged local sensitivity of spreads to scheduled fundamental surprises. An alternative is constructed from debt service, reserves, maturity structure, and primary-balance forecasts. Contemporaneous spreads are excluded from the proxy because they would create a mechanical narrative-by-spread interaction.

For scheduled fiscal release e , estimate

$$\text{Takeup}_{ie} = a_i + \tau_e + \pi_1 \text{Surprise}_{ie} + \pi_2 \text{Surprise}_{ie} B_{i,e-1} + \varepsilon_{ie}, \quad (10)$$

$$\Delta s_{ie} = b_1 \text{Surprise}_{ie} + b_2 \hat{\varepsilon}_{ie} + b_3 \hat{\varepsilon}_{ie} B_{i,e-1} + u_{ie}. \quad (11)$$

Eurostat publishes dated calendars for government deficit and debt releases (Eurostat, 2026). Residual take-up remains endogenous to unobserved news. The release design improves timing and conditioning. It does not create random narrative exposure.

6.3 Supportive variation and interpretation

Foreign media diffusion can be summarized by $N_{it}^F = \sum_{j \neq i} w_{ij,0} N_{jt}$ using predetermined language and media-overlap weights. Common investors, bank exposures, trade, and regional news threaten the exclusion restriction. This design is supportive unless those channels are directly absorbed

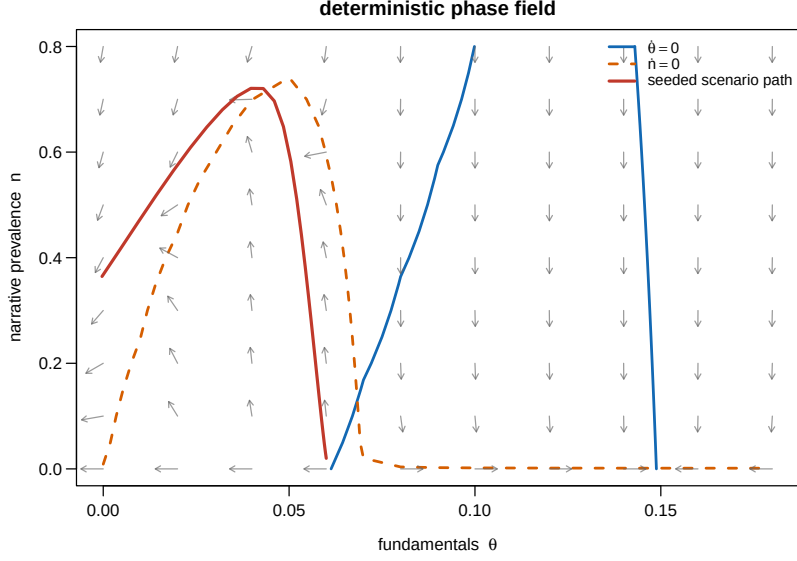


Figure 4: Deterministic phase field. The solid blue curve is the fundamentals nullcline and the dashed orange curve is the narrative nullcline. Arrows show the local vector field. The red curve is the seeded scenario path. A nullcline is a local zero-drift locus, not a global basin boundary.

and the remaining overlap shifts coverage without shifting repayment capacity.

The OMT episode is an interrupted event study. It changed tail-risk insurance, conditionality, and the expected policy regime. It is useful for comparing narrative volume, diffusion speed, sovereign yields, and CDS around a major backstop. It is not a pure test of narrative deletion. Lead-lag local projections can test whether text tends to precede prices near the estimated boundary. Granger timing is a dynamic fingerprint, not causal identification.

Table 2 separates restrictions of the model from scenario magnitudes. Quantitative conversion of text coverage into investor prevalence requires estimation of λ_i . Without that step, the market regressions remain reduced form.

7 Conclusion

The default boundary gives a precise restriction on narrative pricing. A fixed pessimistic shift has a direct price effect equal to the normal probability mass between two default thresholds. That effect has one peak. Equilibrium feedback preserves one peak only under an additional derivative condition. The condition is stated analytically and verified on the quantitative state space.

Narrative adoption is tied to decision value through an attention-cost distribution. The prevalence equation is continuous in time. Media diffusion remains an observed proxy linked to latent investor prevalence through a measurement equation. These distinctions keep a text count from being interpreted as a wealth share without evidence.

The numerical results are scenarios. At the stated inputs, ten percentage points of latent

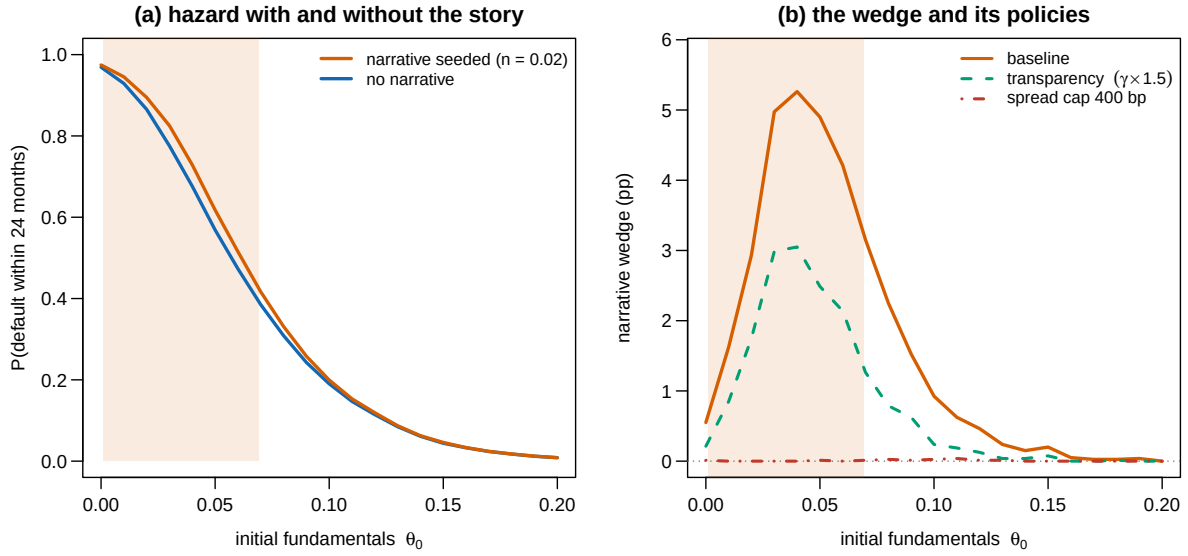


Figure 5: The narrative crisis wedge. (a) 24-month default probability by initial fundamentals, seeded against unseeded, common shocks. (b) The wedge under baseline, transparency, and a 400 bp cap (cap wedge measured against its own unseeded baseline).

prevalence can raise spreads by 230 basis points near the boundary, and a seed can raise the two-year default probability by 5.3 percentage points. The phase portrait is needed to establish whether a local drift movement becomes a global default path. A spread cap removes the modeled marginal effect only where it binds. OMT changed several policy objects at once and cannot identify narrative deletion by itself.



Figure 6: The price of waiting, at the wedge peak: 24-month default probability as a function of the month in which the policy arrives.

Table 2: Predictions and implementation.

Prediction	Model	magni-	Null	Estimation
Narrative pricing is hump-shaped in predetermined boundary proximity	230 vs 2.3 bp scenario		uniform effect	flexible local projections
Scheduled surprises have stronger media take-up near the boundary	Proposition 2		uniform take-up	release windows
Residual take-up has state-dependent price effects	equations (10) and (11)		no interaction	release windows
Media diffusion and latent prevalence obey a measurement system	equation (8)		direct equality	state space
Backstop events change several channels jointly	cap scenario		pure deletion	interrupted event study
Text-price timing changes across boundary bins	dynamic fingerprint	finger-	uniform timing	local projections

Table 3: Data and implementation.

Dataset	Coverage	Freq.	Access	Status	Use
GDELT 2.0 GKG and Mentions (GDELT Project, 2015)	global, 2015–	15 min	public	media text	core diffusion panel
Factiva or LexisNexis	historical press	daily	licensed	full text	historical validation
Consensus forecasts	major sovereigns	monthly	licensed	observed	licensed panel
Bond and CDS spreads	liquid sovereigns	daily	licensed	observed	licensed panel
Fiscal releases (Eurostat, EU and national 2026)		event	public	observed	surprise windows
Public debt (World Bank, reporting economies 2026)		quarterly	public	observed	cash-flow inputs
ECB Data Portal (Euro-pean Central Bank, 2026)	euro area	daily	public	sovereign yields	market outcomes

A Proofs

A.1 Proof of Lemma 1

Write $F(s) = (1 - \mathcal{R})\bar{p}(\theta, s, n)$. Then

$$F'(s) = (1 - \mathcal{R})\frac{\chi}{\sigma}[(1 - n)\phi(z) + n\phi(z + \Delta)] \leq \frac{(1 - \mathcal{R})\chi}{\sigma\sqrt{2\pi}} < 1.$$

The map sends $[0, 1 - \mathcal{R}]$ into itself and is a contraction. Monotonicity follows by implicit differentiation because the denominator $1 - F'(s^*)$ is positive. The same differentiation gives the multiplier. A cap replaces F by $\min\{F, \bar{s}\}$, which is Lipschitz with the same contraction modulus. Failure of the sufficient bound does not prove multiplicity. It permits it when the fixed-point map crosses the 45-degree line at the required tangencies. ■

A.2 Proof of Theorem 1

Let $G(z) = \Phi(z + \Delta) - \Phi(z) = \int_z^{z+\Delta} \phi(u) du$. Since $\phi'(u) = -u\phi(u)$,

$$G'(z) = \phi(z + \Delta) - \phi(z).$$

For $\Delta > 0$, equality $\phi(z + \Delta) = \phi(z)$ holds iff $(z + \Delta)^2 = z^2$, hence iff $z = -\Delta/2$. If $z < -\Delta/2$, then $|z + \Delta| < |z|$ and $G'(z) > 0$. If $z > -\Delta/2$, then $|z + \Delta| > |z|$ and $G'(z) < 0$. Thus G is single-peaked at $-\Delta/2$ and converges to zero in both tails. Differentiating the pricing fixed point gives

$$\frac{dz^*}{d\theta} = -\frac{\rho}{\sigma} M(\theta, n) < 0,$$

so the single-peaked profile in z maps into a single-peaked profile in fundamentals. At $z = -\Delta/2$, symmetry gives $G = 2\Phi(\Delta/2) - 1$. ■

A.3 Proof of Proposition 1

For fixed n , write the equilibrium price derivative as

$$v(z, n) = (1 - \mathcal{R})\frac{G(z)}{1 - ad(z, n)}.$$

The contraction condition makes the denominator positive. Logarithmic differentiation gives

$$\frac{\partial \log v(z, n)}{\partial z} = Q(z, n).$$

If $Q_z < 0$, the log derivative is strictly decreasing and has at most one zero. If it takes a positive and a negative value, that zero exists and is the unique maximum. Since $z^*(\theta, n)$ is strictly decreasing in θ , the same uniqueness holds in fundamentals. The reported numerical margin

evaluates the analytic derivative

$$Q_z = (\log G)'' + \frac{a\{d''(1 - ad) + a(d')^2\}}{(1 - ad)^2}$$

on the stated grid. ■

A.4 Proof of Proposition 2

Proposition 1 makes $v(\theta, 0)$ single-peaked with zero tails. Strict monotonicity of H_c preserves that ordering. Multiplication by $\bar{m}A/\gamma$ preserves it again. A continuous single-peaked function has interval superlevel sets. The set at level one is nonempty exactly when its maximum weakly exceeds one. Increasing A raises the function pointwise and increasing γ lowers it pointwise, which gives the set inclusions. ■

A.5 Proof of Proposition 3

The implicit function theorem applies at the lower root because $D_\theta > 0$. Lemma 1 gives $s_n^* > 0$. Hence $D_n = -\chi_m s_n^* < 0$ and

$$\hat{\theta}'(n) = -\frac{D_n}{D_\theta} = \frac{\chi_m s_n^*}{D_\theta} > 0.$$

No statement about the global basin follows without restrictions on the second component of the vector field. ■

A.6 Proof of Proposition 4

Near zero prevalence with constant rates, $n(t) = n(0) \exp[(\beta - \gamma)t] + o(n(0))$. Since $\beta - \gamma = \gamma(R_0 - 1)$, a cut from n to $(1 - f)n$ is undone after the stated delay. Raising γ lowers $R_0 = \beta/\gamma$ and lowers $1 - \gamma/\beta$ where $\beta > \gamma$ is fixed. Where a cap binds, the modelled spread is locally constant in n . Assumption 1 then gives zero adoption through decision value. The remaining law is $\dot{n} = -\gamma n$. The fundamentals drift also replaces the uncapped spread with the cap. ■

B The Pricing Block: Details and the Multiplicity Frontier

Spread mapping. The baseline uses linear expected-loss pricing, $s = (1 - \mathcal{R})\bar{p}$. The alternative mapping $s = -\log[1 - \bar{p}(1 - \mathcal{R})]$ is more convex at high default probability. It changes both the multiplier and the equilibrium-hump condition. No robustness claim for that mapping is used in the propositions.

The Calvo frontier. The condition $(1 - \mathcal{R})\chi\phi(0)/\sigma < 1$ is sufficient for contraction. The baseline scenario has maximal gain 0.64. At $\chi = 0.30$ it reaches 0.96. Beyond the contraction region, uniqueness is no longer guaranteed and equilibrium selection must be specified (Calvo, 1988; Aguiar et al., 2022).

Wealth-weighted pricing. Averaging beliefs by wealth presumes infected and susceptible investors hold with equal intensity. Letting the infected also shorten duration or exit produces an effective prevalence exceeding n in the pricing equation and strengthens every result. The text uses the lower-intensity baseline.

C Simulation Details

Monthly parameters use the exact aggregation in equation (2). The pricing fixed point is solved to 10^{-12} on a 226×51 grid over $(\theta, n) \in [-0.10, 0.35] \times [0, 1]$. Simulation uses bilinear interpolation. The continuous-time logistic prevalence equation is advanced by its exact one-month flow with current rates held fixed. Monte Carlo experiments use 8,000 paths of 24 months with common random numbers across configurations. Default is absorbing at $\theta < 0$. Deterministic exhibits remove the fundamentals shock. Policy experiments switch the relevant law in the stated month. All figures and reported scenario values are generated from one JSON result file.

D Parameter Discipline and Sensitivity

Table 4 records the role of each input. Setting $\chi = 0$ removes the debt-service feedback while retaining direct narrative pricing. As loop gain approaches one, the contraction margin becomes the relevant sensitivity statistic.

Table 4: Parameter discipline and sensitivity.

Parameter	Enters at	Units	Source or moment	Range examined	Effect of the range
ρ, σ	eq. (3)	annual, capacity	scenario, data targets	joint scale	scale z and Δ jointly
$\bar{\theta} = 0.15$	eq. (1)	capacity	scenario	safe-tail anchor	anchors the safe tail
$\mathcal{R} = 0.60$	pricing	recovery rate	Cruces and Trebesch (2013)	recovery anchor	scales s^* and the hump
$\chi = 0.20$	eq. (7)	capacity per spread	scenario, cash-flow target	0.10 to 0.30	wedge 0.8 to 10.2 pp, gain 0.96 at the top
$\xi = 0.03$	eq. (4)	capacity	scenario, belief target	via Δ	hump height and width
$\gamma = 0.30$	eq. (5)	per month	scenario, episode target	$\times 1.5$	wedge 5.3 to 3.0 pp
$R_0 \text{ peak} = 4$	eq. (6)	unitless	scenario, diffusion target	$A \in \{0.25, 1, 1.5\}$	zone width 0.001 to [0.001, 0.081]
$n_0 = 0.02$	initial condition	latent share	scenario, onset target	timing experiments	delay formula only
cap = 400 bp	policy	spread	OMT-scale instrument	month 1 to 13	hazard 48.5 to 65.4 percent
loop gain	Lemma 1	unitless	implied	2χ one	crosses multiplicity, App. B

Notes: recovery is externally anchored. The remaining non-implied values are scenario inputs. Estimation requires the data moments in Section 6.

E Data and Measurement Protocol

Sources. GDELT 2.0 supplies the public media-diffusion core (GDELT Project, 2015). Factiva or LexisNexis supplies licensed historical full text. Consensus forecasts measure scheduled-release surprises. Eurostat supplies dated deficit and debt releases (Eurostat, 2026). World Bank QPSD and debt-office cash-flow schedules discipline the feedback block (World Bank, 2026).

Classification. A dictionary layer supplies high recall. A frozen multilingual classifier requires the structured claim defined in Section 6. A blinded, double-coded sample supplies inter-rater reliability and out-of-sample error rates by country and language.

Construction. The text index is the share of country-specific sovereign coverage carrying a classified narrative. Media diffusion uses first-appearance timestamps and new-outlet adoption within story clusters. Equation (8) separates this observed index from latent investor prevalence. Survey beliefs can validate the loading where comparable questions exist (Andre et al., 2026).

F Extensions

Cross-border story networks. Let prevalence in country i respond to a reach-weighted sum of foreign prevalences. This structure can generate correlated spread movements among susceptible countries and provides the model counterpart to the supportive foreign-diffusion design (Longstaff et al., 2011).

Endogenous amplification. Attention to sovereign stories rises with volatility, making A procyclical: the zone widens exactly when fundamentals deteriorate, an accelerant with the same signature as a rising multiplier and a natural explanation for why crisis eras feel narrative-dense.

Immunity and waves. Investors who abandoned a story once may resist it: an SIRS structure with slow waning of immunity produces refractory periods after outbreaks and recurring waves at business-cycle frequency, testable in long text archives: the same story striking the same country twice should spread slower the second time, controlling for the boundary.

Heterogeneous stories. A mixture over bite sizes ξ concentrates the priced composition toward severe stories as the boundary nears, since the hump’s window widens with Δ : near the boundary, discourse should grow and darken, a compositional prediction the classification layer can test.

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